



Module 4: Signals, Circuits, Antennas & Safety

Equipment · Digital Modes · Antennas · Feed Lines · Safety
13 exam questions — 37% of the test

Part 1: Station Equipment

T7A & T7B · 2 exam questions



Inside a Transceiver

- Oscillator — generates the signal at your frequency
- Mixer — combines signals to shift frequency (tuning)
- RF Amplifier — boosts signal for transmission
- Audio Amplifier — boosts received audio for the speaker
- Filter — passes desired frequencies, blocks unwanted
- Modulator — adds your voice to the signal
- Demodulator — extracts voice from received signal



SWR — Standing Wave Ratio

- Measures how well your antenna matches your radio
- 1:1 = perfect match (all power to antenna)
- 1.5:1 = excellent
- 2:1 = good, acceptable
- 3:1 = marginal — radio may reduce power
- ∞ :1 = no antenna or dead short — all power reflected
- SWR meter goes in the feed line between radio and antenna



Test Equipment & Troubleshooting

- Dummy load — 50 Ω non-radiating load; test without transmitting
- Wattmeter — measures RF power output
- Multimeter — measures V, I, R (the basic tool)
- Solder: use ROSIN-CORE (never acid-core for electronics)
- No power? Check supply, check fuse
- High SWR? Check antenna connection, coax, wrong band
- Distorted audio? Mic gain, compression, bad mic element

Part 2: Modulation Modes

T8A · 1 exam question



Voice Modes

FM (Frequency Modulation)

- Varies the frequency of the signal
- ~16 kHz bandwidth (widest)
- Used on VHF/UHF repeaters and simplex
- Clear audio, but uses the most spectrum
- Most common mode for Technicians

SSB & CW

- SSB: single sideband — efficient, ~3 kHz
- USB above 10 MHz, LSB below (convention)
- Used for HF phone and weak-signal VHF
- CW: Morse code — on/off keying, ~150 Hz
- Narrowest bandwidth = longest range
- AM: ~6 kHz, old-style HF — rarely used

Part 3: Digital Modes

T8B, T8C, T8D · 3 exam questions



FT8 — The Hottest Mode in Ham Radio

- Uses WSJT-X software + radio + computer
- Contacts take ~15 seconds (computer handles it)
- Works with signals FAR below what you can hear
- Perfect for Technicians on 10m and 6m
- Can work the world with low power and a simple antenna
- This is what gets new hams hooked on HF



APRS, Winlink, and More

- APRS — automatic position reporting with GPS + radio
- Digipeaters relay APRS packets; see stations on aprs.fi
- Winlink — email over radio; critical for emergency comms
- PSK31 — keyboard-to-keyboard text; narrow bandwidth
- Packet radio — data networking over radio
- Mesh networking — IP broadband over amateur radio
- DMR, D-STAR, System Fusion — digital voice modes



Amateur Satellites

- FM repeaters in space — talk through them with an HT
- Doppler shift: freq shifts as satellite passes overhead
- Moving toward you = higher frequency, away = lower
- Track with apps or online satellite trackers
- Technicians can use amateur satellites on VHF/UHF
- No license upgrade needed — just point and talk

Part 4: Antennas

T9A · 1 exam question



Antenna Types

Omnidirectional

- Vertical — radiates in all directions
- 5/8-wave vertical — VHF favorite, low-angle gain
- Rubber duck — HT convenience, poor efficiency
- Mobile whip — mag-mount or permanent
- Ground plane — vertical with radials

Directional

- Dipole — bidirectional (figure-8 pattern)
- Yagi (beam) — high gain, narrow beam
- Director + driven element + reflector
- Front-to-back ratio = rejection from behind
- Great for DXing — point at the target



Antenna Concepts

- Dipole length (ft) = $468 \div \text{frequency (MHz)}$
- 146 MHz dipole = $468 \div 146 \approx 3.2$ feet total
- Gain — directional antenna concentrates signal (dBi/dBd)
- Polarization — match antenna to signal (vertical FM, horizontal SSB)
- A rubber duck is convenient but terrible — external antenna is a huge upgrade
- Antenna is the single most important part of your station

Part 5: Feed Lines & Connectors

T9B · 1 exam question



Coaxial Cable & Connectors

- Use 50-ohm coax (RG-58, RG-8X, RG-213, LMR-400)
- NEVER use 75-ohm TV cable (RG-59/RG-6) — wrong impedance
- Loss increases with frequency AND with length
- LMR-400 = lowest loss; RG-58 = highest loss (but fine for short runs)
- PL-259 / SO-239 = most common connector
- BNC = HTs and test equipment; SMA = modern HTs; Type N = UHF/microwave
- Balun: connects balanced antenna (dipole) to unbalanced feed (coax)
- Weatherproof outdoor connections — water kills coax

Part 6: Safety

T0A, T0B, T0C · 3 exam questions



Electrical Safety

- As little as 30V can be lethal
- Power supplies hold charge in capacitors — even after off
- Always measure before touching a circuit
- Replace fuses with SAME rating — never higher
- GFCI outlets in the shack
- Ground your station: single-point ground to ground rod
- Disconnect antennas during lightning storms



Antenna & RF Safety

- POWER LINES are the #1 killer in amateur radio
- Never install antennas where they can contact power lines
- Never climb a tower alone
- MPE = Maximum Permissible Exposure (FCC RF safety limit)
- Duty cycle: FM = ~100%, SSB = ~20%, CW = ~40%
- Higher duty cycle = more exposure = stricter limits
- Control exposure: less power, more distance, antennas high
- FCC requires RF exposure evaluation for your station



Key Numbers — Module 4

Equipment & Modes

- SWR 1:1 = perfect match
- Dummy load = 50Ω, non-radiating
- CW = narrowest bandwidth (~150 Hz)
- FM = widest common voice (~16 kHz)
- USB above 10 MHz, LSB below
- FT8 = WSJT-X, 15-sec contacts
- APRS = position reporting

Antennas & Safety

- Dipole: $468 \div \text{MHz} = \text{feet}$
- Standard coax: 50 ohms
- PL-259 = most common connector
- LMR-400 = least loss
- Balun: balanced ↔ unbalanced
- Power lines = #1 danger
- Rosin-core solder only

Checkpoint Quiz

10 questions from T7, T8, T9, T0



What does an SWR of 1:1 indicate?

- A. Maximum power reflected
- B. Antenna not connected
- C. Perfect impedance match
- D. Feed line is shorted



What is a dummy load used for?

- A. Increase range
- B. Test without radiating
- C. Match impedance
- D. Filter signals



Which emission has the narrowest bandwidth?

- A. FM
- B. SSB
- C. AM
- D. CW



What software is used for FT8?

- A. Echolink
- B. WSJT-X
- C. Ham Radio Deluxe
- D. Winlink



What is APRS?

- A. Encryption system
- B. Automatic position reporting
- C. Amplifier control
- D. Antenna tuning



Approximate total length of a 146 MHz half-wave dipole?

- A. 10 feet
- B. 6 feet
- C. 3 feet
- D. 1 foot



Which coax has the least signal loss?

- A. RG-58
- B. RG-59
- C. RG-8
- D. LMR-400



Most important antenna installation safety rule?

- A. Always use a balun
- B. Never near power lines
- C. Use thickest coax
- D. Mount as low as possible



What solder for electronics?

- A. Acid-core
- B. Silver
- C. Rosin-core
- D. Plumber's



What does MPE stand for?

- A. Maximum Power Emission
- B. Maximum Permissible Exposure
- C. Minimum Power Efficiency
- D. Modulated Power Evaluation

Before Module 5

Your final assignment



Your Assignment

- Drill HamStudy: T7A–T7D, T8A–T8D, T9A–T9B, T0A–T0C
- This is the most sections — start NOW, don't wait
- Target: 80%+ on every section
- Module 5 = Capstone: full practice exam + review + VE exam
- Bring your photo ID and your FRN if you have one
- You're one session away from your license
- The hard work is done — now you prove it